

Day 88 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Indian Economy & TN Dev. Admin. (6): India's Unicorn Startup Ecosystem 2026

READING MATERIAL

Covers India's startup 'unicorn' (companies valued at \$1+ billion) ecosystem and its newest entrants in 2026.

Q: How many unicorn startups does India have as of mid-2026, and where does this rank globally?

A: 87 unicorns — ranking 3rd globally, behind the USA (1,196 unicorns, 1st) and China (248 unicorns, 2nd).

Q: Which Indian city leads in number of unicorn startups, and what are the next 2 cities?

A: Bengaluru leads with 55 unicorns, followed by Mumbai (22) and Gurugram (21).

Q: Which sector has produced the most Indian unicorns?

A: Consumer (60 unicorns), followed by Enterprise Applications (41) and Retail (38).

Q: Name 2026's first 3 new Indian unicorns, in order, and their respective fields.

A: 1. Juspay (January 2026) — fintech/payments. 2. KreditBee (April 2026) — fintech/lending. 3. Skyroot Aerospace (7 May 2026) — private spaceflight/rocket launch services.

MOCK TEST (WITH ANSWERS)

1. As of mid-2026, India has how many unicorn startups?

- A. 67
- B. 77
- C. 87 ✓**
- D. 97

Explanation: 87. [medium, web-research]

2. India ranks where globally in total number of unicorn startups?

- A. 1st
- B. 2nd
- C. 3rd ✓**
- D. 4th

Explanation: 3rd, behind the USA and China. [medium, web-research]

3. Which Indian city has the most unicorn startups?

- A. Mumbai
- B. Bengaluru ✓**
- C. Gurugram
- D. Delhi

Explanation: Bengaluru (55 unicorns). [easy, web-research]

4. Skyroot Aerospace, which became one of India's newest unicorns on 7 May 2026, operates in which sector?

- A. E-commerce
- B. Private spaceflight/rocket launch services ✓**
- C. Food delivery
- D. Edtech

Explanation: Private spaceflight/rocket launch services. [medium, web-research]

5. Which sector has produced the most Indian unicorns?

- A. Consumer ✓**
- B. Enterprise Applications
- C. Retail
- D. Healthcare

Explanation: Consumer (60 unicorns). [hard, web-research]

Part B – Aptitude & Mental Ability: Area

READING MATERIAL

Area is 1 of 10 named Aptitude sub-skills. Real exam evidence shows a strong preference for applied/real-world framing (fencing cost, carpet cost, crossroads, a grazing horse) over bare formula plug-ins — and frequently combines area with a real-world cost-per-unit calculation.

Q: Method — area of a triangle from coordinates: Find the area of the triangle with vertices (0,0), (2,0), (0,2).

A: When two sides lie along the axes (a right triangle at the origin), area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2 \times 2 = 2$ sq. units. (The general coordinate formula $\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$ also works, but recognizing the right-angle-at-origin shortcut is faster.)

Q: Real exam: A rectangular ground 80m $\times$ 60m has two equal-width cross roads (one parallel to each side). Their combined area is 675 sq.m. Find the road width.

A: If width=w, the two roads' combined area = $80w + 60w - w^2$ (subtracting the overlapping square where they cross, counted twice otherwise) = $140w - w^2$. Set equal to 675: $w^2 - 140w + 675 = 0$. Solving gives $w=5$ (the other root, 135, is too large to be physically valid for an 80 $\times$ 60 ground).

Q: Real exam: Find the perimeter of a semicircle of radius 28cm.

A: Perimeter of a semicircle = $\pi r + 2r$ (the curved half plus the straight diameter edge). Using $\pi \approx 22/7$: $(22/7 \times 28) + (2 \times 28) = 88 + 56 = 144$ cm. (Common mistake: forgetting to add the straight diameter edge, which would wrongly give just 88cm.)

Q: Real exam: A horse is tethered by a 14m rope to one corner of a 60m $\times$ 42m rectangular field. How much area is left UNGRAZED?

A: The horse can graze a quarter-circle of radius 14m (since it's tied at a corner, with two field-edges as boundaries): grazed area = $\frac{1}{4} \times \pi \times 14^2 = \frac{1}{4} \times (22/7) \times 196 = 154$ sq.m. Total field area = $60 \times 42 = 2520$ sq.m. Ungrazed = $2520 - 154 = 2366$ sq.m.

Q: Real exam: Area of a rhombus with diagonals 6cm and 8cm?

A: Area of rhombus = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 6 \times 8 = 24$ cm².

Q: Real exam: If fencing a circular park at ₹5/metre costs ₹1,100 total, find the park's radius.

A: Total fencing length (= circumference) = $₹1,100 \div ₹5/\text{m} = 220$ m. Circumference = $2\pi r \rightarrow 220 = 2 \times (22/7) \times r \rightarrow r = 220 \times 7/44 = 35$ m.

Q: Real exam: Find the cost of carpeting a room 13m long, 9m wide, using a carpet 75cm wide, at Rs.8 per metre (of carpet length).

A: Room area = $13 \times 9 = 117$ m². Carpet width = 0.75m, so length of carpet needed = Room area $\div$ carpet width = $117/0.75 = 156$ m. Cost = $156 \times \text{Rs.}8 = \text{Rs.}1,248$.

Q: Real exam (related, surface area of a 3D shape): The curved surface area of a right circular cylinder of height 7cm is 66 cm². Find its diameter.

A: CSA = $2\pi rh \rightarrow 66 = 2 \times (22/7) \times r \times 7 = 44r \rightarrow r = 1.5$ cm $\rightarrow$ diameter = 3cm.

MOCK TEST (WITH ANSWERS)

1. Find the area of the triangle with vertices (0,0), (5,0), (0,4).

A. 10 sq. units ✓

B. 20 sq. units

C. 9 sq. units

D. 18 sq. units

Explanation: Right triangle at origin: area = $\frac{1}{2} \times 5 \times 4 = 10$ sq. units. [easy, authored]

2. Find the perimeter of a semicircle of radius 21cm (use $\pi \approx 22/7$).

A. 66 cm

B. 108 cm ✓

C. 132 cm

D. 96 cm

Explanation: Perimeter = $\pi r + 2r = (22/7 \times 21) + (2 \times 21) = 66 + 42 = 108$ cm. [medium, authored]

3. Find the area of a rhombus with diagonals 10cm and 12cm.

A. 60 cm² ✓

B. 120 cm²

C. 30 cm²

D. 100 cm^2

Explanation: Area = $\frac{1}{2} \times 10 \times 12 = 60\text{ cm}^2$. [easy, authored]

4. Fencing a circular garden at ₹4 per metre costs ₹880 in total. Find the garden's radius (use $\pi \approx 22/7$).

A. 28 m

B. 35 m ✓

C. 22 m

D. 14 m

Explanation: Circumference = $880/4 = 220\text{m}$. $r = 220 \times 7 / (2 \times 22) = 1540/44 = 35\text{m}$. [medium, authored]

5. Find the cost of carpeting a room 10m long and 8m wide with a carpet 80cm wide, at Rs.10 per metre.

A. Rs.800

B. Rs.1,000 ✓

C. Rs.1,200

D. Rs.960

Explanation: Room area = 80 m^2 . Carpet length needed = $80/0.8 = 100\text{m}$. Cost = $100 \times \text{Rs.}10 = \text{Rs.}1,000$. [medium, authored]

6. A horse is tethered by a 7m rope to a corner of a $20\text{m} \times 15\text{m}$ rectangular field. Find the grazed area (use $\pi \approx 22/7$).

A. 38.5 m^2 ✓

B. 44 m^2

C. 49 m^2

D. 77 m^2

Explanation: Quarter circle of radius 7m: $\frac{1}{4} \times (22/7) \times 49 = 38.5\text{ m}^2$. [medium, authored]

Part C – General English: Poem: "Sea Fever" — John Masefield

READING MATERIAL

A well-known poem about the pull of the sea, from Masefield's 1902 collection Salt-Water Ballads. Real coaching-site evidence shows heavy figure-of-speech testing per line, plus structure (anaphora, rhyme scheme) — both covered below.

Q: What is the structure and central theme of "Sea Fever"?

A: Three stanzas, each opening with the same declaration 'I must go down to the seas again' (anaphora — repetition at the start of successive lines/stanzas), expressing an irresistible compulsion to return to sailing. Stanza 1 focuses on navigation and sensory details of sailing; stanza 2 on the wildness/freedom of the sea; stanza 3 contrasts the adventurous 'vagrant gypsy life' with the peace found after a long voyage.

Q: Identify the figure of speech: "where the wind's like a whetted knife."

A: Simile — directly compares the wind's sharpness to a 'whetted' (sharpened) knife using 'like'.

Q: Identify the figure of speech: "the wheel's kick and the wind's song and the white sail's shaking."

A: Personification — the ship's wheel 'kicks', the wind 'sings', giving human-like actions to non-human/inanimate things.

Q: Identify the figure of speech: "And the flung spray and the blown spume, and the sea-gulls crying."

A: Alliteration — repeated initial sounds ('flung'/'spray', 'blown'/'spume').

Q: Identify the figure of speech: "And quiet sleep and a sweet dream when the long trick's over."

A: Alliteration — repeated 's' sounds ('sleep'/'sweet').

Q: What is the rhyme scheme of the opening stanza, and what word rhymes with 'shaking'?

A: AABB rhyme scheme. 'Shaking' rhymes with 'breaking' (as in 'a tall ship and a star to steer her by ... And the wheel's kick and the wind's song and the white sail's shaking, And a grey mist on the sea's face, and a grey dawn breaking').

Q: What does "the sea's face" personify, and what other phrases use the same device?

A: It gives the sea a human-like 'face'. Other personification-style phrases include 'the wind's song' and 'call of the running tide' — the sea and wind are repeatedly given human qualities throughout the poem.

Q: Which collection did "Sea Fever" first appear in, and when?

A: Salt-Water Ballads, John Masefield's 1902 poetry collection.

MOCK TEST (WITH ANSWERS)

1. "Sea Fever" begins every stanza with the same line, "I must go down to the seas again." This repetition at the start of successive stanzas is called:

- A. Simile
- B. Anaphora ✓**
- C. Onomatopoeia
- D. Oxymoron

Explanation: Anaphora — repetition of a phrase at the start of successive lines/stanzas. [medium, web-research]

2. "Where the wind's like a whetted knife" uses which figure of speech?

- A. Metaphor
- B. Simile ✓**
- C. Personification
- D. Hyperbole

Explanation: Simile — uses 'like' to compare the wind to a sharpened knife. [easy, web-research]

3. "The wheel's kick and the wind's song" is an example of:

- A. Simile
- B. Personification ✓**
- C. Onomatopoeia
- D. Alliteration

Explanation: Personification — the wheel 'kicks' and the wind 'sings'. [medium, web-research]

4. "And the flung spray and the blown spume" demonstrates which figure of speech?

A. Alliteration ✓

B. Simile

C. Metaphor

D. Hyperbole

Explanation: Alliteration — repeated initial consonant sounds. [easy, web-research]

5. The rhyme scheme of the opening stanza of "Sea Fever" is:

A. ABAB

B. ABBA

C. AABB ✓

D. ABCB

Explanation: AABB. [medium, web-research]

6. Which word rhymes with "shaking" in the opening stanza?

A. Star

B. Song

C. Breaking ✓

D. Face

Explanation: 'Shaking' rhymes with 'breaking'. [medium, web-research]

7. "Sea Fever" first appeared in which 1902 collection by John Masefield?

A. Salt-Water Ballads ✓

B. Collected Poems

C. The Everlasting Mercy

D. Reynard the Fox

Explanation: Salt-Water Ballads (1902). [hard, web-research]

8. The phrase "the sea's face" gives the sea a human-like quality. This is an example of:

A. Simile

B. Personification ✓

C. Hyperbole

D. Onomatopoeia

Explanation: Personification. [easy, web-research]